STUDYNEST | CLASS 11 PHYSICS

# MARKING SCHEME AND MODEL ANSWERS

## Chapter 11: Thermodynamics

**Maximum Marks: 40 | Suggested time: 60 minutes**

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Thermal equilibrium is transitive.

Microscopic energy of molecules and their interactions.

Q=DeltaU+W.

Q=0.

Temperature constant.

No.

pV=nRT.

1-T_c/T_h.

Zero boundary work.

500 J.

50%.

Zero.

100 J.

Isothermal: T constant; adiabatic: Q=0.

600 J.

40%.

W=pDeltaV=2000 J.

Second law requires rejected heat or other entropy increase.

Reversible: ideal quasi-static frictionless; irreversible: friction or finite-temperature heat flow.

Carnot limit 50%; actual efficiency 33.3%; Qout=1000 J.